

Araz Hamayeli Mehrabani

Mechanical Engineer | Product Development, CAE & Structural Dynamics

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Available for full-time employment from March 2027



Professional Profile

Mechanical engineer with nearly six years of industrial experience developing machinery, renewable-energy mechanisms and mechanical systems from concept and analysis through manufacturing and installation. Experienced in 3D CAD, structural and dynamic CAE, vibration-system development, component selection, production documentation and cross-functional manufacturing support. Currently completing an M.Sc. in Wind Energy Engineering, adding OpenFAST simulation and Python automation to an industrial product-development background.

Core Engineering Expertise

Product development	Requirements definition, concept development, detailed design, design for manufacturing, installation support
CAD & documentation	CATIA V5, SOLIDWORKS, Siemens NX, manufacturing drawings, bills of materials, SAP-based product data
CAE & dynamics	ANSYS Workbench, structural analysis, modal analysis, harmonic response, vibration and resonance assessment
Machinery & mechanisms	Vibrating machinery, motor and excitation-system selection, solar tracking and automated cleaning mechanisms
Programming & simulation	Python, MATLAB/Simulink, OpenFAST, Power BI, engineering workflow automation
Project execution	Manufacturing and assembly support, technical coordination, design revisions, supplier and shop-floor communication

Professional Experience

Development Engineer - Solar Tracking Systems & Automated Mechanisms <i>Paydar Solar-Energy Systems</i>	01/2022–02/2023 <i>Tehran</i>
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- Designed photovoltaic tracking structures, mounting components and mechanical assemblies in SOLIDWORKS for reliable manufacture and installation.
- Used ANSYS structural analyses to refine components, contributing to an approximately 10% reduction in production scrap and raw-material waste.
- Developed an automated panel-cleaning mechanism using electric motors, guide rails and wheels for controlled movement across inclined panel surfaces.
- Delivered the cleaning mechanism from concept and detailed design through manufacturing, assembly and successful on-site installation.
- Prepared manufacturing drawings, bills of materials and component specifications, and maintained engineering data in SAP-based systems.

Mechanical Engineer - Machinery Development & Structural Dynamics <i>TehranRigzar Mining & Mineral-Processing Equipment</i>	03/2017–12/2021 <i>Tehran</i>
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- Co-developed a high-frequency vibrating screen based on laboratory screening studies and defined system requirements.
- Helped replace the legacy crankshaft-based concept with a simpler vibration-motor-driven architecture designed for improved reliability and maintainability.
- Created complete machine and subsystem assemblies in CATIA V5, including structural components, interfaces and manufacturing drawings.
- Performed modal and harmonic-response analyses in ANSYS to evaluate natural frequencies, resonance behaviour and dynamic response.
- Calculated operating requirements with the engineering team and supported vibration-motor selection for the final machine configuration.
- Supported fabrication and assembly, resolving design and fit issues during production.
- Assisted with customer-site installation of the completed machine, which was successfully put into operation.

Current Technical Specialisation

M.Sc. Wind Energy Engineering - Flensburg University of Applied Sciences

03/2023–present

- **Master's thesis:** Developing an OpenFAST-based method to estimate rotor-effective wind speed from measurable turbine-response signals.
- Automating simulation preparation, signal processing and estimator evaluation using Python.
- Analysed aeroelastic loads and structural response for a 20 MW offshore monopile wind turbine using OpenFAST, Python and MATLAB.
- Developed Python utilities for OpenFAST case generation, batch execution, monitoring and output organisation.
- Completed a wind-farm planning study in windPRO and QGIS, covering layout, energy yield, wakes, noise and shadow flicker.

Selected Engineering Projects

Autonomous Underwater Welding Robot - Motion Control

M.Sc. thesis project

- Developed and implemented a MATLAB/Simulink motion-control system enabling precise movement and positioning of an autonomous underwater welding robot.

Hand Prosthesis & Additive Manufacturing

Volunteer project

- Designed and prototyped mechanical components for a hand prosthesis using 3D printing, supporting iterative development from digital model to manufactured parts.
- Worked with 3D-printer hardware and production settings to manufacture and refine prototype components.

Education

M.Sc. Wind Energy Engineering - Current grade average: 1.9

03/2023–present

Flensburg University of Applied Sciences, Flensburg, Germany

Expected graduation: February 2027

M.Sc. Mechanical Engineering

09/2014–01/2017

Hakim Sabzevari University, Sabzevar

Control systems and robotics

B.Sc. Mechanical Engineering

09/2008–06/2013

Islamic Azad University, Tehran

Mechanical design and vibration analysis

Designed a high-frequency screening machine in CATIA V5 and evaluated its structural and vibration behaviour using ANSYS.

Selected Training & Certifications

Python programming HarvardX

ANSYS IEEE training

SOLIDWORKS Technical and Vocational Training Organization

CATIA V5 Amirkabir University of Technology

Languages

English IELTS overall score 7.0

German B1 (self-assessed); preparing for certification